



**Politecnico
di Torino**

Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

Group 1

Daniele Cataldo, Francesco Daraia, Michele Merla

April 6, 2025

Contents

1	Introduction	1
1.1	Theory	1
2	Methodology	2
2.1	Equipment	2
2.2	Methodology	3
2.2.1	Calibration	3
2.2.2	Setup of the software	3
2.2.3	Measurement	4
2.2.4	Data analysis	4

CHAPTER 1

Introduction

The objective of the laboratory is to use a Vector Network Analyzer (VNA) to measure the scattering parameters of an unknown RF device. Through the S-parameters of the device under test (DUT), one can understand its functionality and behavior.

1.1 Theory

Consider a N-port device, conceived as a black box. The scattering matrix elements (S-parameters) relate the incident wave with the transmitted and reflected ones at each port and also provide an insight of the behavior in frequency of the network.

Let us call a_i the incident wave at i -th port and b_j the transmitted/reflected wave at j -th port. The scattering parameter S_{ij} is defined as

$$S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_k=0, k \neq j}$$

where i is the input port and j is the output port. The condition $a_k = 0$ is obtained by terminating the k -th port with a matched load.

Let us now consider a 2-port device.

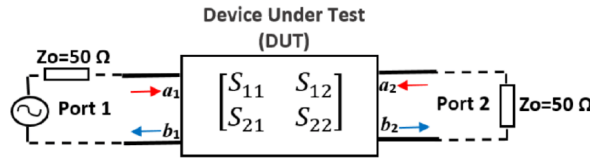


Figure 1.1

The relations between the incident and reflected waves can be expressed by a system of equations:

$$\begin{cases} b_1 = S_{11}a_1 + S_{12}a_2 \\ b_2 = S_{21}a_1 + S_{22}a_2 \end{cases}$$

which admits the following matricial representation:

$$S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}.$$

This matrix is known as **scattering matrix** and its elements are the scattering parameters.

CHAPTER 2

Methodology

2.1 Equipment

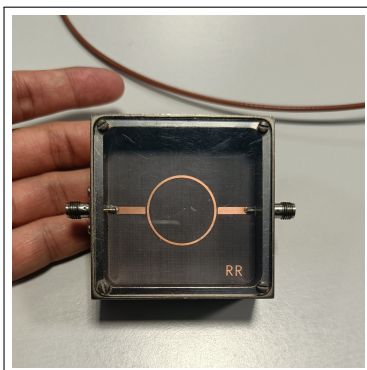
- Vector Network Analyzer (VNA) - Model: Keysight P9371A;



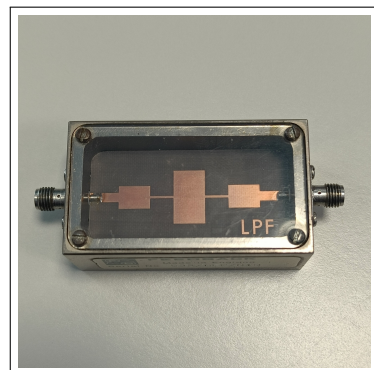
- RF cables and connectors;
- Devices under test (DUTs):
 - Unknown RF device 1 - Serial Number: 532/AMP/018;
 - Unknown RF device 2 - Serial Number: 532/RR/018;
 - Unknown RF device 3 - Serial Number: 532/LPF/013;
 - Unknown RF device 4 - Serial Number: 532/MI/013;
 - Unknown RF device 5 - Serial Number: 532/PIN/018.



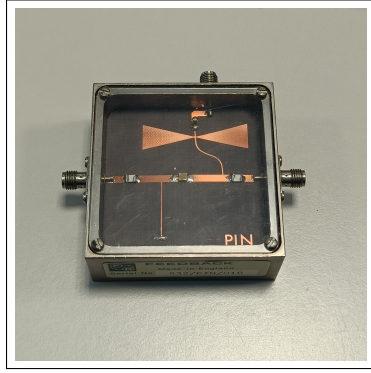
(a) AMP



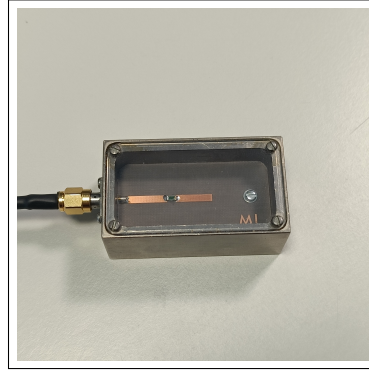
(b) RR



(c) LPF



(a) PIN



(b) MI

- Computer with Keysight Network Analyzer software;
- Electronic Calibration Kit - Model: Keysight N7551A;



2.2 Methodology

2.2.1 Calibration

Before measuring any of the parameters, we calibrated the device using an *Electronic Calibration Kit*. We connected the E-Cal kit to the VNA through an USB port located on the rear panel and the port 1 and 2 of the VNA to port A and B of the calibration kit. Despite performing 3 calibrations, the S_{12} parameter measured was noisy, making it difficult to identify maxima and minima. Therefore, under suggestion of the professor we implemented an IF bandwidth filter to clear the signal. This reduced the noise but increased the measurement time.

2.2.2 Setup of the software

Once the calibration was terminated, we configured some parameters of the Network Analyzer.

- **Traces:** we added 4 traces, one for each scattering parameters needed. S_{11} and S_{22} were displayed on one window while S_{12} and S_{21} in another. In the case of the antenna we used only 2 traces.
- **Start and end frequencies:** we set the start frequency to 10 MHz and the stop frequency at 6.5 GHz.
- **Number of points:** For each trace we set 1001 points.
- **Number of measurements averaged:** we used an average on 8 measurements.

2.2.3 Measurement

We connected each DUT to the VNA through coaxial cables and waited some time to let the IF bandwidth filter reduce the noise. The scattering parameters were displayed on the Network Analyzer software. After that, we proceeded to save the results in a *.s2p* file for two-ports devices and in *.s1p* for the one-port DUT.

2.2.4 Data analysis

S_{11} represents the portion of the input signal at port 1 that is reflected back to port 1. Similarly, S_{22} quantifies the reflection at port 2. S_{22} indicates how much of the signal entering port 1 is transmitted to port 2, while S_{12} describes the transmission from port 2 to port 1.

All these parameters are complex and therefore can be written as the product of the amplitude and the phase. In this case, the phase helps us understand if and how the signal is shifted in time (delayed) when it crosses the device under test.

AMP

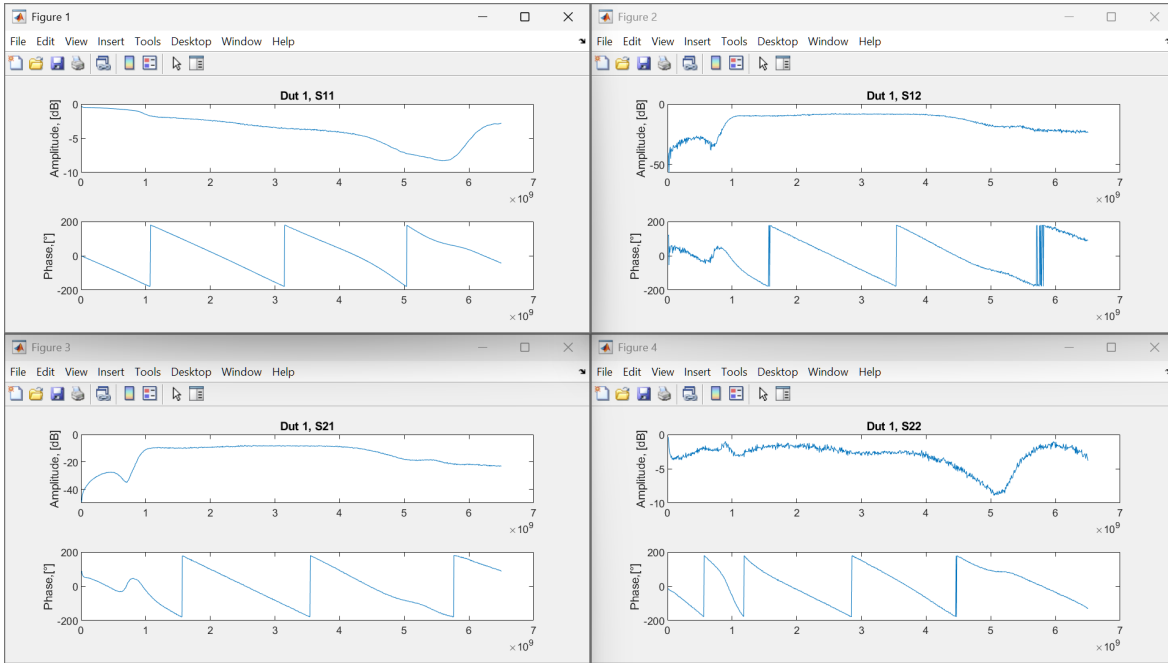


Figure 2.3

The device is **reciprocal**. The transmission increases as the frequency increases. Once the wave enters the DUT, it is almost totally transmitted through the other port. It is interesting the behavior of the phases that shows a linear decrease with wraparounds. This suggests that the DUT can work as a phase shifter because, depending on the working frequency, it is possible to select the phase shift of the transmitted wave.

RR

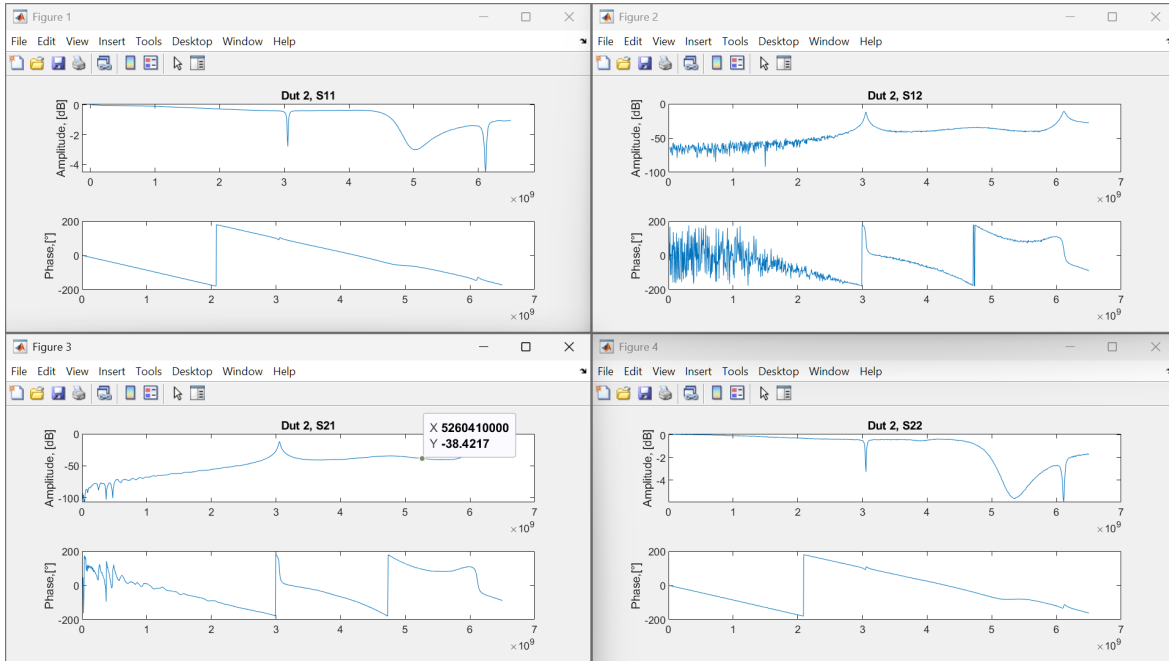


Figure 2.4

The device is **reciprocal**, this is evinced from the graphs of S_{12} and S_{21} , which are almost identical. Therefore, its behavior is independent from the input port. At low frequencies, up until 3 GHz, the device completely reflects the input signals. At around 3 GHz and 6 GHz, the majority of the wave is transmitted, acting as a Notch filter. Also around 5 GHz, it presents another minimum of reflection, spanning a greater range of frequencies. For this reason, we suppose that this DUT is a resonator.

LPF

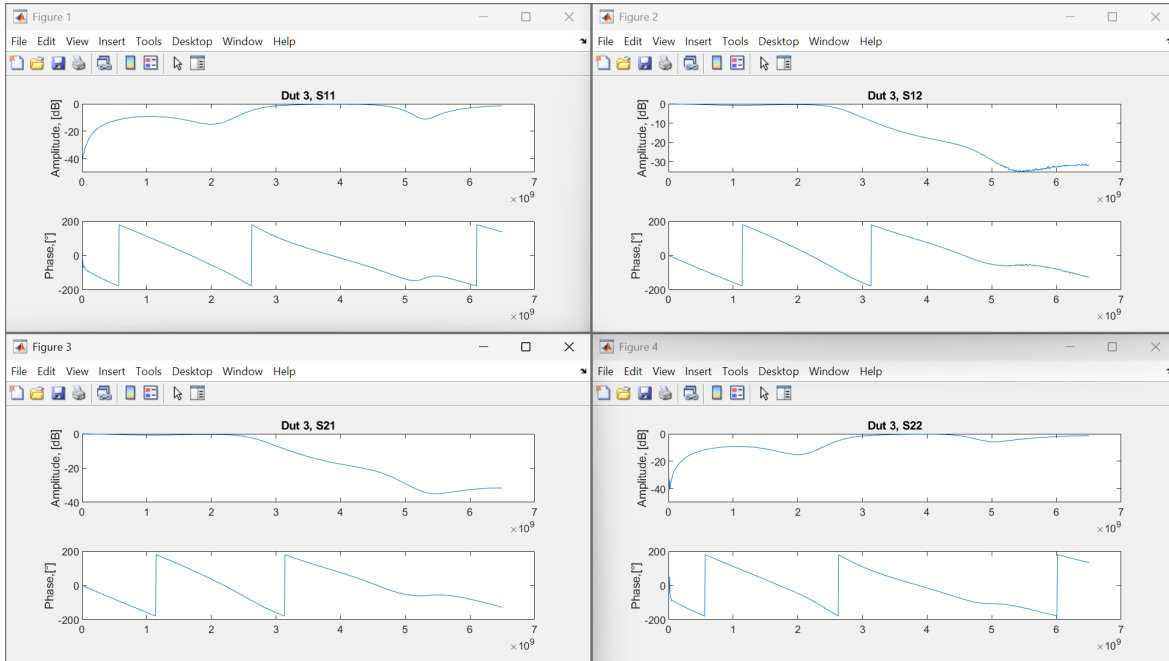


Figure 2.5

The device is **reciprocal**. It reflects every signal starting from 3 GHz. Whereas, low-frequency signals up until 2 GHz are mostly transmitted, as it is shown by S_{12} and S_{21} .

At 5 GHz, the reflection coefficient exhibits a minimum both at port 1 and 2. So, the signal enters the device, but a part is reflected back. However, it does not manage to be transmitted through the other port.

The DUT behaves as a **low-pass filter**.

MI

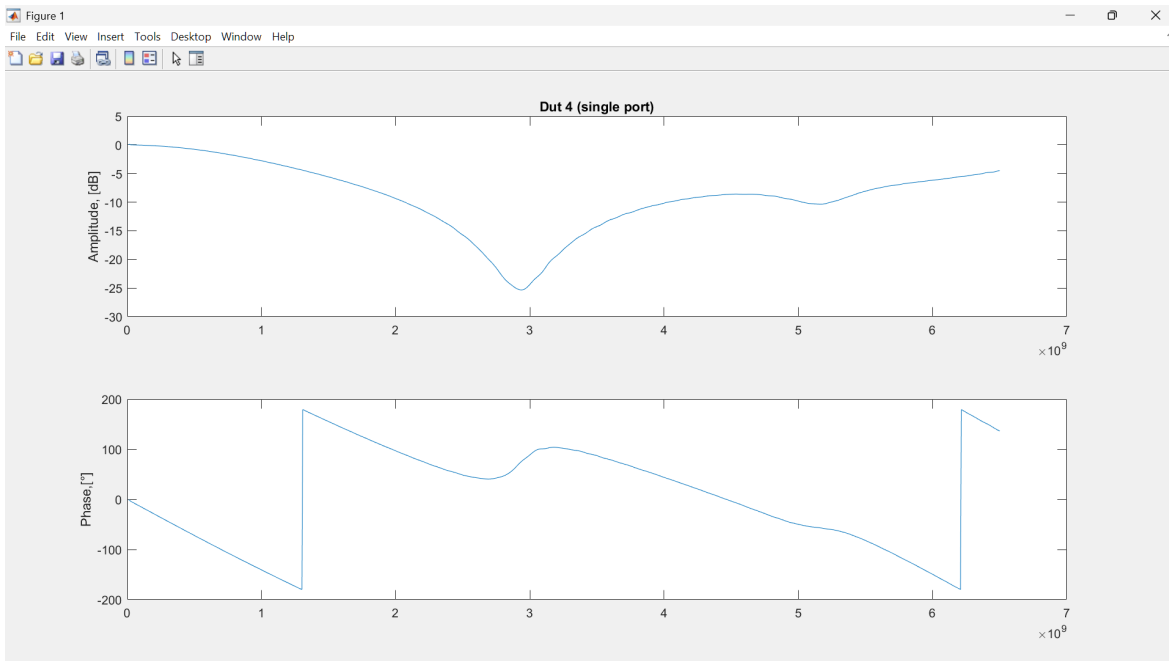


Figure 2.6

The device presents a minimum of reflection at 3 GHz, with a phase shift of about 90° . Although the primary transmission occurs at this frequency, other frequency components are also transmitted through the device, but with noticeable attenuation.

PIN

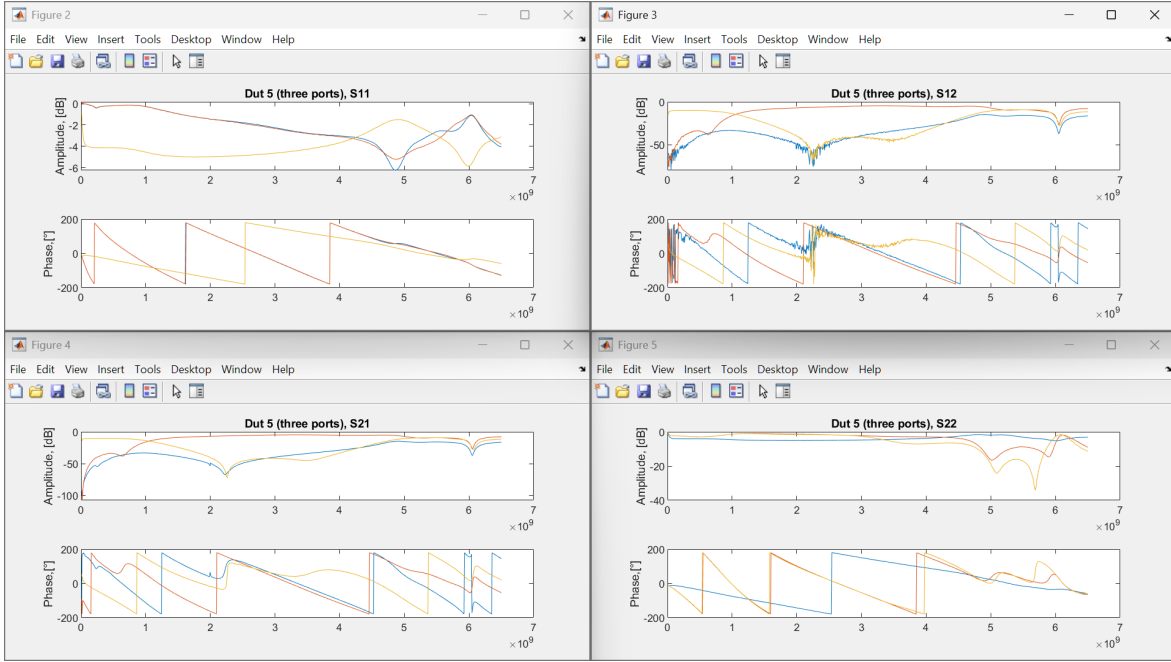


Figure 2.7: The three plots of the S_{ij} resulting from terminating the k-th port. In this way, one can study a three-port device using a two-port VNA.

We can consider every plot as a separated two port device, obtaining an estimation of the behaviour of the DUT.

- **Red:** the device has a peak in trasmission at around 5 GHz, with around half of the wave passing through. In the range 0.6-6 GHz, the wave propagates through the DUT with no attenuation. It acts as a frequency selector.
- **Blue:** as regards S_{11} the behavior is approximately the same as the one plotted in red. On the other hand, the wave is not trasmitted as can be seen with S_{12} and S_{21} . S_{22} instead suggest that the wave is mostly reflected.
- **Yellow:** S_{11} and S_{22} behave similar to the red combination of ports, S_{21} and S_{12} instead behave similar to the blue combination.

What we can state is that for some range of frequencies, it is more convenient to select one port as the input one with respect to the other two. The same applies for the output port. We are not able to deduce the kind of DUT we are dealing with, based on the available data.